

Ayush Kumar Pandey

B.Tech Information Science Engineering — Presidency University, Bangalore

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Bangalore, Karnataka, India

Professional Summary

B.Tech Information Science Engineering student (6th Semester, CGPA: 8.55/10.0) at Presidency University, Bangalore, with 2+ years of hands-on experience in Java full-stack development, machine learning, computer vision, image processing, and Android application development.

Proficient in AI-assisted software development – skilled at prompt engineering, designing and orchestrating AI agent workflows, and architecting production systems with the help of AI tools. Consistently delivers projects faster and more efficiently by leveraging AI agents intelligently, with a focus on minimising token consumption and maximising output quality. The majority of production projects built to date have been designed and developed using AI agent collaboration and optimised agentic workflows.

Currently pursuing a strong interest in Large Language Models (LLMs), Generative AI, and Deep Learning – with an NPTEL certification in LLMs completed. Seeking a Software Engineering internship (starting 7th Semester) with a forward-thinking Bangalore-based organisation.

Education

Presidency University

Bangalore, Karnataka

Bachelor of Technology – Information Science and Engineering — CGPA: 8.55 / 10.0 Sept 2023 – June 2027

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Artificial Intelligence & Machine Learning, Software Engineering, Cloud Computing, Database Management Systems, Operating Systems, Computer Networks, Software Testing & Quality Assurance

Roll Number: 20231ISE0057

Technical Skills

Programming Languages: Java, Python, JavaScript, SQL, HTML, CSS

Java Ecosystem: Spring Boot, JavaFX, Java Collections Framework, Design Patterns (Factory, Strategy, Observer), OOP, Android Studio, Android SDK

Web & Backend: Flask, React, Vite, TypeScript, Tailwind CSS, RESTful APIs, PySide6

AI Agents & Prompt Engineering: Agentic Workflow Design, Prompt Optimisation, Token-Efficient Pipeline Design, System Architecture with AI, Multi-Agent Orchestration, n8n Workflow Automation, UiPath RPA

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLOv8, ArcFace, InsightFace, ONNX Runtime, FAISS, CNN, Transfer Learning, HDBSCAN, MTCNN, DeepFace

Image Processing & Computer Vision: OpenCV, Tesseract OCR, Feature Extraction, Object Detection, Face Recognition, Edge Deployment

Cloud & DevOps: AWS (EC2, S3, VPC), Docker, Git, GitHub, Linux, CI/CD

Databases: MySQL, SQLite, FAISS (Vector DB), Database Design, Data Modelling

Automation & Testing: Selenium WebDriver, BeautifulSoup, Pandas, Unit Testing, Software QA

Embedded & IoT: Raspberry Pi, Arduino, IoT Sensor Integration, Edge Deployment

Tools & Methodologies: PyInstaller, NSIS, JIRA, Agile, Scrum

Experience

AI-Assisted Full-Stack Developer (Project-Based)

2023 – Present

Independent — Java, Python, Android Studio, Flask, React, AI Agents, Prompt Engineering

Bangalore, India

- Architected and shipped 12+ production-ready projects by effectively leveraging AI agents and advanced prompt engineering – designing optimised agentic workflows and system architectures that maximise development speed while minimising token consumption and cost.

- Demonstrated consistent ability to communicate precisely with AI systems: decomposing complex engineering problems into well-structured prompts, orchestrating multi-step agent workflows, and iterating efficiently to production-quality outputs.
- Designed full-stack Java desktop and Android applications applying MVC architecture, OOP principles, and Gang-of-Four design patterns (Factory, Strategy, Observer).
- Developed 15+ modular mathematical tools with a real-time 2D/3D graphing engine, achieving 40% faster feature development through clean OOP and AI-assisted architecture design.

Machine Learning & Computer Vision Developer (Project-Based)

2023 – Present

Independent — TensorFlow, PyTorch, OpenCV, ArcFace, FAISS, YOLOv8

Bangalore, India

- Built and deployed end-to-end ML pipelines for real-world applications: facial recognition (90–94% accuracy), ASL hand sign classification (94%), and vehicle license plate detection (88%).
- Architected FaceSort, a fully offline Windows desktop application using ArcFace 512-d ONNX embeddings and FAISS cosine similarity search with confidence-based auto-assign and human review workflow.
- Used AI agents throughout the development cycle – from architecture planning to debugging and code review – delivering production-quality software significantly faster than traditional solo development.
- Engineered MVC-based multi-threaded systems using ProcessPoolExecutor for CPU-bound ML tasks and ThreadPool for disk I/O; packaged as a production Windows installer via PyInstaller and NSIS.

Image Processing & Computer Vision Developer (Project-Based)

2023 – 2024

Independent — OpenCV, Tesseract OCR, Raspberry Pi, TensorFlow

Bangalore, India

- Designed automated vehicle license plate detection system achieving 88% accuracy with sub-2-second latency and 95% reduction in manual data entry via cloud logging.
- Deployed computer vision models on Raspberry Pi edge hardware with real-time sub-3-second inference.
- Built an IoT server room monitor with 24/7 distributed sensing (temperature, humidity, smoke) and real-time cloud alerts.

Android Mobile Application Developer (Project-Based)

2025

Independent — Android Studio, Java, REST APIs, Material Design

Bangalore, India

- Developed Android mobile applications integrating REST APIs, following Material Design guidelines and Android development best practices.
- Implemented activity lifecycle management, fragment navigation, and asynchronous API calls for smooth user experiences.

AI Workflow & Automation Developer (Project-Based)

2025

Independent — Python, Selenium, n8n, UiPath RPA, AI Agents

Bangalore, India

- Automated ERP timetable scraping with 6-XPath fallback selector logic for AJAX-rendered DOM; exported structured Excel attendance reports.
- Automated repetitive ERP evaluation forms dynamically detecting iframes and handling AJAX; reduced per-subject completion from minutes to seconds.
- Designed and implemented n8n-based automation workflows orchestrated with AI agents for efficient, low-overhead task pipelines.

Projects

FaceSort – Intelligent Offline Photo Management Desktop Application | PySide6, ArcFace, ONNX Runtime, FAISS, S

- Re-architected a Flask/React web prototype into a fully offline Windows desktop app with premium dark UI; system architecture designed collaboratively with AI agents to ensure optimal module separation and performance.
- ArcFace 512-d ONNX embeddings + FAISS cosine similarity: auto-assigns faces above 0.72, queues 0.55–0.72 for human review, creates new profiles below 0.55.
- MVC architecture with ProcessPoolExecutor (CPU-bound ML) and ThreadPool (disk I/O); single-click NSIS installer via PyInstaller.
- **GitHub:** github.com/Ayush-2401/FaceSort-ML-Project

FaceSort v1.0 – Full-Stack Web Application | Flask, React, Vite, TypeScript, Tailwind CSS, MTCNN, DeepFace, HDBSCAN

- RESTful Flask API + React SPA for automated photo organisation through facial recognition; processed 1000+ images at 90% accuracy, reducing manual sorting time by 85%.

- Integrated MTCNN face detection, DeepFace embeddings, FAISS vector search, and HDBSCAN clustering into a single end-to-end ML pipeline.

Advanced Scientific Calculator Platform | *Java, JavaFX, Design Patterns (Factory, Strategy, Observer)* 2024

- 15+ mathematical tools with custom real-time 2D/3D graphing engine; modular OOP architecture delivers 40% faster feature development. Architecture planned with AI agent assistance.

Hand Sign to Alphabet Recognition System | *Python, TensorFlow, CNN, OpenCV* 2024

- 94% real-time ASL alphabet accuracy using custom CNN with data augmentation and transfer learning; live webcam inference.

Attendance Automation & ERP Scraping Tool | *Python, Selenium, BeautifulSoup, Pandas, openpyxl* 2025

- Automated ERP timetable scraping with 6-XPath fallback for AJAX DOM; filtered cancelled classes via CSS inspection; normalised course codes with regex; exported structured Excel reports.

Faculty Evaluation Bot – ERP Automation | *Python, Selenium, AI-assisted workflow design* 2025

- Automated repetitive Linways ERP evaluation forms: detects iframes, handles AJAX, auto-fills all dropdowns; reduces per-subject time from minutes to seconds.

Vehicle License Plate Logger & IoT Server Room Monitor | *Raspberry Pi, OpenCV, Tesseract OCR, Arduino, IoT Sensing*

- 88% plate detection accuracy, sub-2s latency, 95% reduction in manual data entry; 24/7 IoT sensing (temperature, humidity, smoke) with real-time cloud alerts.

Certifications & Online Courses

Introduction to Large Language Models 2026

NPTEL – IIT/IISC National Programme on Technology Enhanced Learning

- Covered transformer architecture, LLM pre-training, fine-tuning, prompt engineering, and practical LLM application development.
- **Certificate:** NOC26CS88S75020090604410485

Fundamentals of Virtual and Augmented Reality Systems 2026

NPTEL – IIT/IISC National Programme on Technology Enhanced Learning

- Covered VR/AR hardware, rendering pipelines, spatial computing, interaction design, and real-time 3D systems.
- **Certificate:** NOC26CS03S125020181504410485

Achievements

Shipped 12+ production projects by mastering AI agent collaboration and prompt engineering – delivering faster, more efficient, and token-optimised development workflows.

Maintained 8.55 CGPA while independently building production-ready systems across ML, Java full-stack, Android, computer vision, and automation.

Sole developer for FaceSort Desktop v2.0: complete product lifecycle ownership from AI-assisted architecture design and ML pipeline development to UI implementation and Windows installer packaging.

Delivered 88–94% accuracy ML systems with 80–95% effort reduction across all independent projects.

Completed two NPTEL certifications in LLMs and VR/AR, demonstrating commitment to continuous learning beyond curriculum.

Areas of Interest

Core: Software Engineering, Java Full-Stack Development, Android Development, System Design, AI-Assisted Development

AI/ML (Active): Deep Learning, Computer Vision, NLP, Agentic AI | *Interest in: Large Language Models, Generative AI, RAG Systems*

Emerging: Multi-Agent Systems, Edge AI, Applied AI for Bangalore tech ecosystem

Additional Information

Location: Bangalore, Karnataka – open to on-site, hybrid, and remote internship opportunities

Internship Availability: 7th Semester (from approx. January 2026)

Languages: English (Professional), Hindi (Native)

Open Source: Active GitHub contributor – github.com/Ayush-2401